1. First Fit

Advantages

- Allocation is very fast.
- It allocates a process to the first suitable partition.
- It requires less memory scanning.
- It is simple and easy to implement.

Disadvantages

- Memory utilization is not always efficient.
- Large partitions may become occupied by small processes.
- Suitable partitions may not be available for larger processes later.
- It may cause internal fragmentation.

Exam Line

First Fit is fast and easy to implement, but it may lead to poor memory utilization and internal fragmentation.

2. Best Fit

Advantages

- It selects the smallest suitable partition for a process.
- Memory wastage is comparatively lower.
- It provides better memory utilization.
- It increases the possibility of allocating more processes.

Disadvantages

- It requires scanning all partitions before allocation.
- Allocation takes more time.
- It is comparatively more complex to implement.
- It has higher searching overhead.

Exam Line

Best Fit provides better memory utilization and less memory wastage, but it requires scanning all partitions and takes more time.

3. Worst Fit

Advantages

- It allocates a process to the largest available partition.
- By using the largest partition, medium-sized partitions may remain available for future processes.
- Its allocation logic is simple.

Disadvantages

- Large partitions become occupied quickly.
- Memory utilization may be poor.
- It requires scanning all partitions.
- It may fail to allocate a larger number of processes.
- It may cause higher internal fragmentation.

Exam Line

Worst Fit allocates the largest partition first, but it often results in poor memory utilization and higher memory wastage.

Best Fit is the Best Placement Algorithm

Best Fit is considered the best placement algorithm because it allocates each process to the smallest suitable memory partition. As a result, large partitions remain available for larger processes, which improves memory utilization and reduces memory wastage. Therefore, more processes can be allocated successfully than with First Fit or Worst Fit.

Although Best Fit requires scanning all memory partitions and takes more time, it provides the most efficient use of memory. Hence, Best Fit is generally considered the best placement algorithm.

Key Reasons

- Selects the **smallest suitable partition** for each process.
- **Reduces memory wastage.**
- **Improves memory utilization.**

- **Keeps larger partitions available** for larger processes.
- **Increases the chance of allocating more processes.**

Very Short Exam Answer (2 Marks)

Best Fit is the best placement algorithm because it always selects the smallest suitable partition for a process. This reduces memory wastage, improves memory utilization, keeps larger partitions available for larger processes, and increases the possibility of allocating more processes successfully.